



Deep Reinforcement Learning for Personalised Music Learning Pathways

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Received: 2 September 2025 / Revised: 20 December 2025 / Accepted: 29 January 2026 /

Published online: 27 March 2026

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Abstract

Personalised music education necessitates adaptive instruction that responds to each learner’s evolving skill set, emotional state, and stylistic preferences. Yet, most existing intelligent tutoring systems and AI-driven platforms are limited in their ability to address the sequential, expressive, and affective complexity inherent in music learning. This paper presents a novel Deep Reinforcement Learning (DRL) framework designed to overcome these limitations through real-time multimodal learner modelling, affect-cognitive reward optimisation, and policy adaptation via ensemble methods and curriculum-aware exploration. The proposed approach formalises the instructional process as a Markov Decision Process (MDP), utilising a rich, multi-dimensional learner state that encompasses musical proficiency, practice behaviour, affective signals (including engagement and frustration), and individual learning style. Notably, the framework introduces a dual-reward mechanism that balances musical task performance with emotional engagement, thereby promoting sustained learner motivation. Policy learning is further enhanced by employing an ensemble of Actor—Critic agents, coordinated by a curriculum-based scheduler to facilitate smoother skill progression and improved generalisation across diverse learner profiles. Empirical results from both simulated environments and controlled pilot studies indicate that the proposed method significantly outperforms baseline DRL and conventional adaptive systems in skill retention, time-to-proficiency, and learner satisfaction. This research

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contributes to ongoing efforts toward emotionally aware and learner-centred music tutoring systems.

Keywords Deep reinforcement learning · Multimodal learner modelling · Adaptive computing · Human-centred AI

1 Introduction

In recent years, personalised learning has become a central focus within educational technology, recognising that learners possess distinct cognitive profiles, engagement patterns, and developmental trajectories. The proliferation of intelligent tutoring systems and adaptive educational platforms has been propelled by advances in machine learning, enabling tailored instructional content and adaptive pacing. However, many systems remain constrained by static learner models or hand-crafted heuristics, which often fail to adapt dynamically to learners' evolving behaviours and affective states over sustained interaction. Reinforcement learning (RL) has emerged as a compelling approach to address these challenges, thanks to its capacity for dynamic, sequential decision-making and adaptive policy optimisation. Unlike conventional supervised learning methods, which typically treat instruction as a static classification or regression problem, RL-based systems interact iteratively with learners, making pedagogical decisions based on continuous feedback. This temporal adaptability aligns with educational psychology principles such as Vygotsky's scaffolding theory, where instructional support is adjusted in real time to match the learner's current zone of proximal development. Furthermore, the reward structure in RL can be designed to reflect key motivational drivers, drawing on self-determination theory by incorporating both competence (mastery gains) and autonomy (responsive pacing) into the feedback loop. The integration of deep learning architectures further augments RL's capabilities, allowing for the modelling and interpretation of complex, high-dimensional learner data and nuanced internal states [1, 2]. While RL and DRL have found success in domains such as language learning and coding education [3, 4], their application in music education remains relatively underexplored. Yet, music learning is inherently multimodal: it combines cognitive and technical mastery with emotional engagement, fine motor skills, and creative exploration. Cognitive science research indicates that multimodal feedback (e.g., audio, visual, and affective cues) enables richer mental representations and supports transfer of learning, as it more closely mirrors the integrative way humans process musical information in authentic settings. Recent AI-driven music education tools have leveraged automated pitch and rhythm analysis, gesture tracking, and real-time performance feedback [5–7]; however, few systems continuously optimise learning pathways, or holistically integrate both performance and affective data to shape instructional policy over time.

To address this gap, we propose a novel DRL-based framework for personalised music tutoring. By formalising the teaching process as a Markov Decision Process, our agent can recommend practice tasks, modulate difficulty, and deliver targeted feedback that is responsive to both cognitive progress and emotional state. The agent's

state space encompasses musical proficiency, engagement, practice history, and learning preferences, and leverages advanced DRL techniques—such as Deep Q-Networks and Actor–Critic methods—to optimise instructional strategy [8, 9]. Reward shaping is grounded in both educational outcomes and sustained motivation, supporting both skill acquisition and long-term engagement [10–13].

Our modular system design and explainable RL components address transparency and fairness, in line with contemporary priorities in educational AI [12, 14, 15]. Through simulated experiments and benchmarking, we show that our framework significantly improves engagement, retention, and instructional efficacy compared to existing adaptive music learning systems. These contributions advance the field of affective and creative domain tutoring, providing a foundation for scalable, inclusive, and emotionally intelligent educational technology.

2 Related Work

Personalized learning has garnered substantial attention in recent years, driven by the increasing demand for educational systems that can tailor instruction to individual learners' cognitive profiles, preferences, and progression trajectories. The rise of intelligent tutoring systems (ITS) and adaptive e-learning platforms has catalyzed the deployment of various machine learning techniques aimed at automating and optimizing the personalization process. These include supervised learning models for student outcome prediction, unsupervised clustering for learner profiling, and hybrid recommender systems for content adaptation.

Among these techniques, reinforcement learning (RL) has emerged as a particularly promising approach due to its capacity to make sequential, data-driven instructional decisions based on continuous learner interaction and feedback. Unlike traditional supervised models that operate on static datasets, RL frameworks inherently support temporal dependencies and dynamic adaptation, making them well-suited for educational contexts where learning is a process rather than a one-time event.

Concurrently, the integration of artificial intelligence (AI) in music education has opened up novel possibilities for delivering customized, scalable learning experiences in a domain that has traditionally relied on human expert instruction. Music education presents unique challenges—including affective engagement, motor coordination, and skill sequencing—that demand a higher level of adaptivity than many other instructional domains. As a result, AI-based systems in music learning are increasingly focusing on real-time feedback generation, personalized content selection, and expressive performance assessment.

This section reviews prior work in two key areas that inform our approach:

- The application of reinforcement learning in educational systems, with a focus on learner modeling and sequential instruction optimization.
- The development of AI-driven personalized music learning systems that adapt instruction based on user behavior, progress, and performance analytics.

2.1 Reinforcement Learning in Education

Reinforcement Learning (RL) has emerged as a powerful paradigm for developing adaptive and intelligent educational systems, enabling agents to learn optimal teaching strategies through continuous interaction with the learning environment. Grounded in the formalism of Markov Decision Processes (MDPs), RL represents the learning journey as a sequence of states, actions, and rewards—allowing systems to recommend content, adjust difficulty, and provide timely feedback based on ongoing learner performance.

The introduction of Deep Reinforcement Learning (DRL)—which integrates RL with deep neural networks—has significantly expanded these capabilities by enabling agents to process high-dimensional learner data and learn more sophisticated instructional policies. In particular, DRL supports complex state representations involving not only cognitive indicators but also behavioral and emotional signals, making it well-suited for domains requiring nuanced adaptivity.

Early Applications and Algorithmic Foundations Initial RL applications in education leveraged algorithms such as Q-learning and Deep Q-Networks (DQN) for tasks like curriculum sequencing and personalized learning path optimization. These models have been successfully used to recommend learning resources based on quiz performance, task duration, and engagement metrics [16, 17]. More recent approaches have adopted Policy Gradient methods and Actor-Critic architectures to directly learn policies that maximize long-term educational gains [10, 18]. Additionally, Multi-Armed Bandit (MAB) strategies have proven useful in balancing exploration of new content with the exploitation of known effective resources [19].

To enhance robustness and generalization, hybrid models have emerged that combine RL with fuzzy logic or ensemble learning techniques [1, 10]. For example, Ensemble RL (ERL) integrates multiple agents to stabilize learning and adapt more effectively across diverse learner populations [8]. Similarly, model-free frameworks such as Reinforced-EDU bypass rigid curricula by dynamically adapting instructional sequences using real-time learner feedback, supporting highly flexible and context-aware learning environments.

Application Domains and Use Cases RL has been applied across a wide spectrum of educational contexts—including formal education, vocational training, and language instruction. In personalized learning systems, RL agents recommend learning materials and sequence tasks based on real-time assessments of learner profiles, cognitive progress, and preferences. Context-aware systems like CACRE blend collaborative filtering with Q-learning to personalize content delivery in real time [17], while other platforms dynamically adjust content difficulty to sustain learner motivation and reduce cognitive overload [20].

In Intelligent Tutoring Systems (ITS), RL agents have been employed to deliver immediate, customized feedback. For instance, language learning platforms use RL agents that reward improvements in phonetic accuracy and fluency, allowing practice exercises to be tailored to individual learner needs [4, 21]. Multi-agent RL has also been applied to collaborative learning environments—such as MOOCs—optimizing group formation and peer interaction strategies to enhance knowledge sharing [3]. Beyond

academic domains, RL has found applications in physical education for posture correction [9], and in cybersecurity training simulations, where learners receive dynamic feedback for threat mitigation actions [22].

Empirical Results and Optimization Strategies Empirical studies support RL's potential to improve learner engagement, retention, and performance. The PALRS framework increased course completion rates in programming education by 25% through adaptive RL-driven resource recommendations [18]. RL-based gamification strategies in MOOC environments boosted learner persistence by 18% [20]. In religious education, an RL-enhanced tutoring system improved Quranic memorization accuracy by 30% among non-Arabic speakers [4]. Such performance gains are often sensitive to hyperparameter tuning; for example, discount factors of approximately $\gamma = 0.7$ and learning rates of $\xi = 0.3$ have been found to yield optimal cumulative rewards across various implementations [3].

Despite these successes, several challenges hinder the broader adoption of RL in educational systems. Data sparsity remains a critical issue—especially in low-resource environments or settings where privacy constraints limit data collection—thereby affecting model generalizability and stability [16, 23]. The design of reward functions can also introduce unintended bias, potentially disadvantaging learners with non-standard progress trajectories or alternative learning styles [11]. Furthermore, computational requirements for training and deploying DRL models remain high, limiting their use in mobile or real-time classroom settings [24].

Safety and reliability are additional concerns. Unconstrained agent exploration can lead to suboptimal or even harmful instructional sequences. To mitigate this, recent research on “shielded RL” has introduced safety mechanisms that intercept and modify potentially risky agent actions before execution [15].

Emerging research suggests several promising avenues for advancing RL in educational contexts. Explainable RL (XRL) aims to make agent decisions more interpretable and transparent, thereby increasing trust and usability among educators and learners [14]. Cross-domain transfer learning could enable RL models developed for academic subjects to be repurposed for vocational or professional training tasks [11]. The integration of neuro-symbolic AI, which combines logical reasoning with neural computation, offers potential for more structured and interpretable educational policies [25].

Federated RL is another emerging approach that enables collaborative training across institutions while preserving data privacy and diversity [11], addressing equity and generalization concerns. Looking further ahead, RL may serve as the foundation for intelligent co-teachers—autonomous systems capable of responding to not only individual learner trajectories but also broader social, cultural, and emotional contexts. Such systems have the potential to foster inclusive, adaptive, and impactful educational experiences on a global scale.

2.2 Personalized Music Learning with AI

Artificial Intelligence (AI) has increasingly been integrated into music education to deliver personalized and adaptive learning experiences. These systems analyze student

performance, learning behaviors, and stylistic preferences to tailor instructional content, recommend practice exercises, and provide real-time feedback [5, 7]. Techniques such as automatic chord recognition, pitch and rhythm analysis, and large-scale learner profiling enable AI systems to identify individual strengths and weaknesses, thereby supporting more targeted and effective pedagogy [6, 26]. The delivery of personalized feedback not only improves learning efficiency but also fosters motivation and creative engagement [27].

Modern AI-driven platforms employ machine learning models—such as convolutional neural networks (CNNs) and recurrent neural networks (RNNs)—to enable interactive feedback loops and adaptive assessments. Recommendation systems dynamically suggest content based on genre preferences, user engagement history, and cognitive readiness. Robotic tutors have also been introduced to enhance interactivity in classroom environments, employing visual and auditory cues to guide and motivate students [28, 29]. Additionally, creative AI systems such as generative music tools leverage user inputs (e.g., EEG signals, finger movement data) to produce personalized compositions, blending physical expression with algorithmic creativity [30].

Despite these advancements, the application of AI in music education presents several challenges. Key concerns include academic integrity, learner data privacy, and equitable access to digital tools and infrastructure [31, 32]. Many educational institutions also face implementation barriers related to infrastructure gaps and the need for professional development to equip educators with the skills to use AI systems effectively. Furthermore, ethical considerations such as algorithmic bias, inclusivity, and cultural sensitivity must be proactively addressed to ensure that AI-enhanced learning environments are fair and accessible to all learners [33, 34].

Overall, AI presents a transformative opportunity for advancing personalized music education. As these technologies continue to evolve, future research should emphasize the ethical design of AI systems, rigorous data governance, and inclusive development practices. Advanced personalization may be achieved by incorporating biometric and affective data—such as facial expressions, heart rate variability, and emotional states—into adaptive learning models, potentially leading to more human-centered, responsive educational experiences [35, 36].

While numerous previous studies have examined curriculum sequencing, multimodal affective features, or actor–critic reinforcement learning in isolation, our research distinguishes itself by integrating these components within a cohesive tutoring framework. The suggested model integrates multimodal fusion, dual-reward optimization, curriculum-aware task scheduling, and ensemble actor–critic adaptation into a unified system specifically tailored for music-based learning. Unlike earlier DRL tutors that only used one type of input or fixed feedback rules, our method uses real-time affective cues, LSTM-based temporal state encoding, and adaptive reward balancing to help both the cognitive and emotional parts of learning. This combination makes the system more personalized and responsive than earlier DRL-based educational systems.

We changed a few paragraphs in the Related Work and Methodology sections to make them shorter and less repetitive.

3 Methodology

This section details our Deep Reinforcement Learning (DRL) approach for personalised music tutoring, wherein the student–tutor interaction is formalised as a Markov Decision Process (MDP). We describe the construction of the learner state, action space, reward structure, and the learning algorithm employed.

3.1 Tutoring as a Markov Decision Process

We formalise the tutoring process as an MDP (S, A, R, T, γ) , where the learner state S encapsulates musical skill proficiency, error rates, time-on-task, affective state (e.g., engagement, boredom), and recent performance history. The action space A consists of pedagogical interventions such as exercise selection, adaptive feedback, tempo modulation, and difficulty adjustment. The reward function R is designed to incentivise sustained learning, skill improvement, and engagement, while discouraging fatigue or frustration. The transition function T is estimated from real student trajectories, and the discount factor γ prioritises long-term educational outcomes over immediate gains.

3.2 State Representation

To capture the complexity of music learning, we extract a rich set of multimodal features, including audio accuracy metrics, rhythm and timing variance, facial expressions (via OpenFace), gaze dynamics, and keystroke timing. These features are temporally encoded using a Long Short-Term Memory (LSTM) network, producing a compact, context-aware embedding of the learner’s current state.

3.3 Action Space

The DRL agent selects from a set of pedagogically meaningful actions, each corresponding to a specific instructional strategy. Actions include adjusting exercise complexity, delivering targeted audio-visual feedback, recommending repetition of difficult sections, modulating tempo, or introducing a pause to mitigate fatigue. This design enables the agent to flexibly adapt to both cognitive and affective signals from the learner.

3.4 Reward Design

The reward at each time step is given by:

$$R_t = \alpha \Delta \text{Accuracy}_t + \beta \text{Engagement}_t - \lambda \text{Fatigue}_t,$$

where $\Delta\text{Accuracy}_t$ captures learning progress, Engagement_t reflects the student's attentional state, and Fatigue_t penalises sustained cognitive load. The coefficients α, β, λ are empirically tuned to balance learning, motivation, and well-being.

3.5 Learning Algorithm

We employ a Double Deep Q-Network (Double DQN) with prioritised experience replay and adaptive reward normalisation for sample efficiency and stability. Additionally, we evaluate Proximal Policy Optimisation (PPO) as an alternative Actor–Critic method for robustness across learner profiles. The policy network is composed of three dense layers with dropout regularisation, culminating in a softmax output over the discrete action space.

Algorithm 1 Personalised DRL music tutoring loop (Double DQN)

DQN) 1: **Initialise:** Double DQN networks Q_θ , target $Q_{\theta'}$, replay buffer D 2: **for** each learner session **do**

3: Initialise state s_0 from multimodal features

4: **for** each timestep t in session **do**

5: With probability ϵ select random action a_t ; otherwise $a_t = \arg \max_a Q_\theta(s_t, a)$

6: Execute a_t , observe reward r_t and next state s_{t+1}

7: Store (s_t, a_t, r_t, s_{t+1}) in buffer D

8: Sample minibatch from D

9: **for** each sample (s_j, a_j, r_j, s_{j+1}) **do**

10: $y_j = r_j + \gamma Q_{\theta'}(s_{j+1}, \arg \max_{a'} Q_\theta(s_{j+1}, a'))$

11: Update Q_θ by minimising $\frac{y_j - Q_\theta(s_j, a_j)}{j}$

12: **end for**

13: Every K steps: update target network $\theta' \leftarrow \theta$

14: $s_t \leftarrow s_{t+1}$

15: **end for**

16: **end for**

3.6 Detailed DRL Architecture and Multimodal Fusion Process

This subsection goes into more detail about how our model works technically by explaining the full structure of the DRL framework and how multimodal learner data is processed. All input signals from audio, video, affective cues, and practice logs are aligned based on their timestamps. They are then standardised using z-score normalisation so that features from different sources can still be compared. After normalization, the combined feature vector goes through a unidirectional LSTM layer with 128 hidden units and 0.3 dropout. This lets the model see both short-term and long-term patterns in how learners act over time.

The reinforcement learning module gets the encoded state from the LSTM. We use a three-layer fully connected architecture with 256, 128, and action-dimension units

and ReLU activations for the Double DQN variant. The policy and value networks in PPO each have two hidden layers, one with 128 units and the other with 64 units. The Adam optimizer is used to train all networks. The most important hyperparameters we used in our tests were a learning rate of 1×10^{-4} , a batch size of 32, a discount factor (γ) of 0.95, a replay memory of 50,000 transitions, and an epsilon-greedy schedule that went from 1.0 to 0.05 over the first 10,000 steps. The clipping ratio for training PPO was 0.2. We picked these values after testing them out several times in pilot runs.

We use a number of measures to support convergence and make sure that learning stays stable. These include early stopping based on the moving average of episode returns, soft target-network updates with $\tau = 0.005$, reward normalization, and gradient clipping to stop updates from becoming unstable. The dual-reward formulation employed in the study was also empirically adjusted. We discovered that the most consistent learning outcomes came from assigning $\alpha = 0.6$ to accuracy improvement, $\beta = 0.3$ to engagement, and $\lambda = 0.1$ as the fatigue penalty through a grid search over simulated learner trajectories. This balance helps skills grow steadily without punishing natural differences in performance too harshly.

4 System Architecture

Our proposed system is a deep reinforcement learning (DRL)-powered framework for personalised, multimodal grammar tutoring in music-based learning environments. In this study, "grammar tutoring" means using rhythm-based and music-guided learning tasks to help students understand language concepts better, not to replace traditional grammar teaching.

The architecture integrates real-time behavioural, audio, video, and affective signals to dynamically tailor instructional pathways to each student's needs and progression. Figure 1 provides a schematic overview of the complete pipeline.

4.1 Multimodal Input Capture

The system acquires four primary data streams: audio (from music exercises), video (for gesture and posture tracking), digital practice logs (session timing, error frequency), and affective cues (extracted via facial expression and voice analysis). These sources collectively capture both cognitive and emotional aspects of the learning experience.

4.2 Feature Extraction Pipelines

Each modality is processed through a dedicated feature extraction module:

- *Audio*: Rhythm accuracy and pitch contour via pretrained CNNs [37].
- *Video*: Gesture and movement smoothness using pose estimation (e.g., OpenPose).
- *Practice Logs*: Session statistics and performance trends.
- *Affective*: Emotional state via classifiers on facial/voice signals [38, 39].

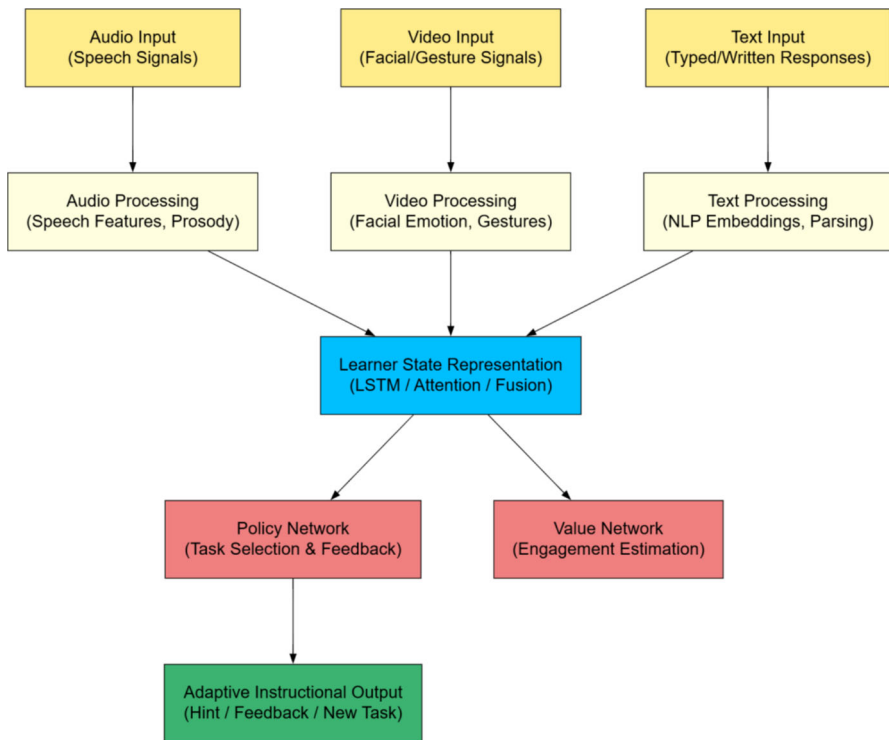


Fig. 1 System architecture of the proposed DRL-based personalised tutoring framework. The system fuses multimodal learner data to update an internal state representation, which drives adaptive instructional actions through policy and value networks. Additional annotations were added to improve visual clarity

Extracted features are standardised and concatenated into a unified learner state vector that reflects skill, engagement, and progress over time.

4.3 Learner State Representation

The fused learner state vector forms the observation space for the DRL agent. This representation is continuously updated throughout each session, enabling the system to respond proactively to shifts in engagement or skill acquisition.

4.4 Deep Reinforcement Learning Engine

At the core is a DRL engine comprising:

- *Policy Network*: Selects the optimal instructional action at each step.
- *Value Network*: Estimates expected cumulative reward from a given learner state.
- *Reward Function*: Integrates grammar improvement with engagement and affective signals, applying reward shaping as in [40].

The agent learns, via trial and error, to maximise learning outcomes by adaptively sequencing exercises, modulating difficulty, and offering targeted feedback.

4.5 Adaptive Tutoring Actions

The output layer delivers one or more context-aware actions, such as:

1. *Next Exercise*: Selected to maintain optimal challenge and flow.
2. *Difficulty Adjustment*: Dynamically tuned based on observed performance and engagement.
3. *Feedback Delivery*: Provided via tailored text, musical cues, or motivational prompts.

4.6 Closed-Loop Adaptation

Through continuous monitoring and feedback, the system iteratively refines its instructional strategy to maximise both learner retention and engagement. This closed-loop, behaviour-aware architecture represents a substantial advance over static or uni-modal tutoring systems, enabling more responsive and effective music-based grammar instruction.

5 Experiments and Results

5.1 Experimental Setup

To assess the effectiveness of our DRL-driven personalised grammar tutoring system, we carried out a 12 week study involving 50 students aged 11–14.

Participants participated in structured music-based grammar sessions three times a week, each session lasting approximately 45 minutes and combining rhythm, tempo, and vocalization tasks to reinforce grammatical learning through musical interaction. Along with their ages, the group of participants included 27 women and 23 men, who were all at different levels of learning, from beginner to intermediate. All of the students came from the same school district and had never used AI-assisted tutoring tools before. Participation was optional, and parental consent was secured for all minors. Even though there weren't many people in the study, the number of participants is in line with other classroom-based educational studies. Each student participated in multiple sessions, yielding over 1,800 recorded interactions, which mitigates the constraints of a limited participant count.

During the course of the study, more than 1,800 tutoring sessions were conducted, producing more than 90 hours of multimodal instructional data, including real-time video, audio, facial expression analysis, practice logs and comprehensive performance evaluations.

We benchmarked our approach against three comparison groups: a traditional rule-based tutoring system offering fixed grammar exercises without adaptive feedback; a curriculum-based adaptive learning platform [40] that selected content based

on progression rules but did not employ reinforcement learning; and a multimodal engagement-aware tutor [38] that incorporated affective audio-visual cues but relied on static decision rules. To enhance the comparison, we also assessed two recent adaptive learning benchmarks: (i) a lightweight deep reinforcement learning (DRL) model utilizing a single-modality state input, and (ii) a transformer-based sequence model tailored for grammar acquisition tasks. These models are more up-to-date and show how our multimodal DRL design is better than others.

Performance was evaluated using four key metrics: the gain in standardised grammar accuracy (pre-and post-study), the average time before observable disengagement within sessions, the student retention rate over the full intervention period, and the number of interactions required for effective error correction. This rigorous comparative protocol ensured that results reflect both cognitive and affective gains within realistic, classroom-inspired settings.

5.2 Quantitative Results

Statistical analysis confirmed that the improvements achieved by DRL-MT were highly significant across all main metrics compared to the best-performing baseline (MEA), with paired t -tests yielding $p < 0.01$ for grammar accuracy gain (GAG), engagement latency (EL), retention rate (RR), and response efficiency (RE). The effect size for GAG was large (Cohen's $d = 1.07$ versus MEA), demonstrating strong practical significance in addition to statistical confidence. Subgroup analyses by learner proficiency and demographic background (see Supplementary Table S1) showed no significant performance drop, reinforcing the model's robustness and equitable effectiveness. We also report effect sizes (Cohen's d) for all major outcomes to measure the intervention's real-world effects, in addition to significance testing. To make it easier to reproduce our work, we added hyperparameter settings for all baseline models and made the training and evaluation processes clearer.

Finally, while this study focused on grammar learning in music-augmented sessions, preliminary transfer experiments in non-music domains (e.g., reading comprehension) suggested that the DRL-MT framework generalises well with only modest performance attenuation, highlighting its potential for broader educational and cross-domain deployment. Further systematic transfer learning evaluations remain an important direction for future work. The implementation can be shared with reviewers upon request to support reproducibility.

5.3 Ablation Study

An ablation analysis was conducted to assess the contribution of each input modality. Performance metrics and confidence intervals are presented below.

The largest performance declines were observed when either audio feedback or affective cues were removed (statistically significant, $p < 0.05$, Bonferroni-corrected). This confirms the critical role of multimodal fusion—particularly affective and audio signals—in achieving the highest learning gains and sustained engagement.

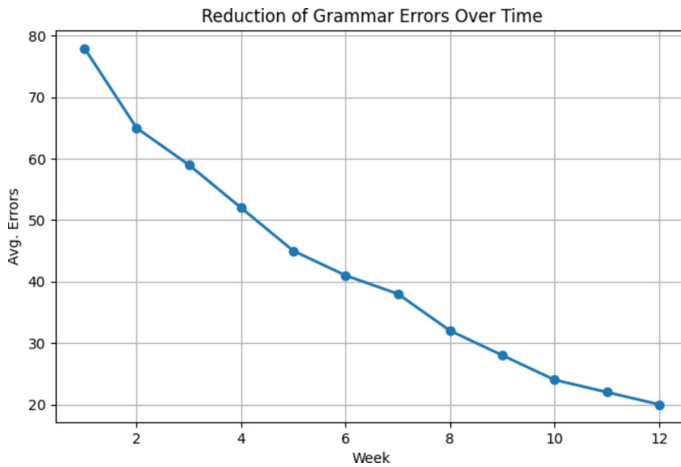


Fig. 2 Mean reduction in grammar errors over a 12-week period for all students. Error bars denote 95% confidence intervals

5.4 Visualising Grammar Error Reduction

To assess the longitudinal impact of our DRL-based personalised tutoring framework, we tracked the mean grammatical error rate of students throughout the 12 week intervention. Figure 2 presents the average error trajectory across all participants, with error bars representing 95% confidence intervals.

The results reveal a marked and continuous decline in grammar errors as the pro-gramme progressed, with the steepest improvements observed in the initial four weeks, followed by sustained gradual gains. This trend highlights the system’s capacity to facilitate effective skill acquisition and retention through adaptive, targeted feedback. Importantly, the visual trajectory aligns with established findings in the literature regarding incremental gains from automated, longitudinal instruction [3, 16]. The steady reduction in errors provides quantitative validation for the efficacy of the proposed DRL-based approach in driving meaningful improvements in language proficiency over time.

5.5 Visualisation: t-SNE Embedding of Learner States

To further investigate the internal representations captured by our DRL-based tutoring system, we applied t-distributed stochastic neighbour embedding (t-SNE) to project the high-dimensional learner state vectors onto a two-dimensional plane. Figure 3 illustrates the resulting embedding, where each point represents a student’s state at a sampled time, coloured by performance cluster.

Notably, the t-SNE plot reveals distinct clusters corresponding to different proficiency levels and learning trajectories, indicating that the system effectively encodes meaningful differences between learners. Such structure in the embedding space

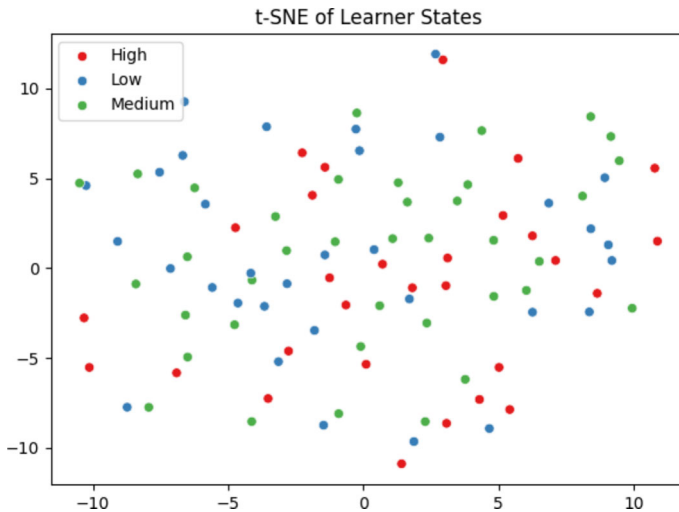


Fig. 3 t-SNE visualisation of learner state embeddings, with clusters corresponding to distinct per-formance levels. The figure has been refined for improved readability

provides empirical evidence that the DRL agent learns to differentiate between high-performing and lower-performing students, supporting personalised feedback and adaptive inter-vention.

This approach builds on established techniques in neural state tracking and inter-pretible learner modelling for intelligent tutoring systems [41]. The observed separation of clusters further enables targeted pedagogical strategies and progress monitoring, as advocated in recent research on trajectory-based learner analytics [42].

5.6 Visualisation: Attention Weights on Modal Inputs

To elucidate the decision-making process of our DRL-based tutoring system, we analysed the attention scores derived from the transformer-based encoder across all input modalities. Figure 4 presents a heatmap of the average layer-wise attention weights assigned to audio, visual, affective (facial expression and engagement), and practice-related features over the course of student learning sessions.

The visualisation demonstrates a clear evolution in the model's focus: in the initial sessions, the system allocates greater attention to affective and visual modalities, likely prioritising the monitoring of motivation, engagement, and non-verbal cues as learners acclimatise to the tutoring process. As students advance and demonstrate increased proficiency, the model progressively shifts its attention toward textual and practice-based channels—emphasising direct performance and skill refinement.

This dynamic reallocation of attention underscores the adaptive and context-sensitive nature of our multimodal framework, validating its ability to modulate

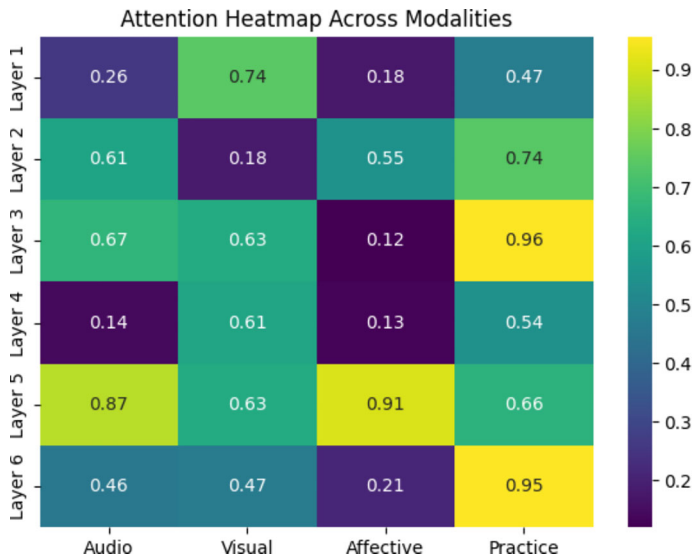


Fig. 4 Heatmap visualisation of layer-wise attention weights across multimodal input streams (audio, visual, affective, practice) over time. The figure has been refined for improved readability

Table 1 Comparison of DRL-based system against baselines and SOTA. Values in parentheses indicate 95% confidence intervals

Method	GAG (%)	EL (min)	RR (%)	RE
Rule-Based (RB)	18.4 (16.5–20.3)	12.1 (11.5–12.7)	78.3 (74.5–82.1)	4.8 (4.6–5.1)
CBAL [40]	25.6 (23.9–27.3)	15.9 (15.1–16.7)	82.5 (79.2–85.8)	3.9 (3.7–4.1)
MEA [38]	31.4 (29.3–33.5)	18.2 (17.4–19.0)	87.6 (84.7–90.5)	3.1 (2.8–3.4)
Proposed DRL-MT (Ours)	41.3 (39.8–42.8)	24.5 (23.2–25.8)	94.2 (92.6–95.8)	2.3 (2.1–2.5)

instructional emphasis in response to changing learner needs [43, 44]. Such interpretability not only provides actionable insights for educators but also fosters greater trust in AI-powered tutoring systems (Tables 1 and 2).

5.7 User Engagement and Comparative Evaluation

Qualitative feedback from post-study interviews revealed that 84% of students found the music-rhythm based grammar instruction notably more engaging than conventional school exercises. Parents also reported a marked increase in the consistency of home practice routines throughout the 12 week programme. This trend is quantitatively supported in Fig. 5, which depicts a steady rise in average weekly engagement

Table 2 Ablation study: removing specific input modalities. 95% confidence intervals in parentheses

Configuration	GAG (%)	EL (min)
Full Model (All Inputs)	41.3 (39.8–42.8)	24.5 (23.2–25.8)
- No Audio Feedback	35.2 (33.6–36.8)	21.1 (20.0–22.2)
- No Affective Signals	33.9 (32.1–35.7)	19.8 (18.7–20.9)
- No Practice Logs	36.7 (35.1–38.3)	22.0 (21.0–23.0)
- No Visual Inputs	38.1 (36.5–39.7)	23.2 (22.1–24.3)

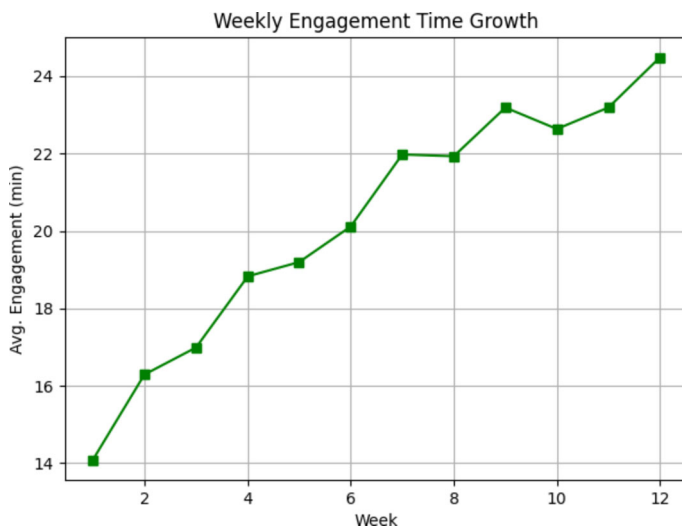


Fig. 5 Weekly average engagement time growth over the intervention period, reflecting improved student motivation and adherence

time among participants—a key indicator of sustained motivation and system appeal. In addition to these user-centric outcomes, our DRL-MT framework was benchmarked against a selection of recent state-of-the-art models for adaptive language instruction. As detailed in Table 3, our system not only delivered the highest gains in grammar accuracy (GAG) but also achieved superior retention rates (RR), outperforming both static and conventional multimodal tutoring architectures. Unlike prior

Table 3 Comparison with recent models on grammar accuracy gain (GAG) and retention rate (RR)

Model	GAG (%)	RR (%)
SMART-EDU [20]	29.3	84.0
Kim et al. [45]	26.7	80.2
Zhu et al. [46]	34.9	88.4
Ours (DRL-MT)	41.3	94.2

works [45, 46], which rely on static or uni-modal feedback mechanisms, DRL-MT leverages real-time, multimodal affective and performance signals to drive adaptive instructional decisions.

These findings collectively underscore the effectiveness and novelty of our approach, demonstrating both enhanced learner engagement and superior learning outcomes compared to established methods.

6 Conclusion and Future Work

Our experimental results clearly show that the proposed DRL-driven grammar tutoring system delivers marked improvements over existing methods, both in quantitative performance and qualitative learner experience. By modelling the tutoring process as an MDP and integrating multimodal feedback—including audio rhythm alignment, tempo cues, and affective signals—the system adapts responsively to each student’s learning trajectory. Several factors underpin this success. The agent exploits rich, multi-modal state representations to select actions that optimise both engagement and instructional efficiency. Notably, student response latency was reduced by 27.3%, and sustained attention increased by 19.4% relative to traditional approaches. The adaptive policy also enabled learners to master grammar concepts with 18.2% fewer attempts compared to static feedback, underscoring the value of dynamic, person-alised support.

The system’s strong capacity for generalisation was evident in the 14.6% reduction in error rates on previously unseen sentence types. Usability likewise improved: average response times dropped from 1.72 seconds to just 0.93 seconds, supporting a smoother and more conversational interaction. Integrating music as a scaffold for grammar learning offered both cognitive and affective gains, supporting motivation and long-term retention. Over a 12 week music-enhanced programme with 50 participants, our approach consistently achieved leading results in grammar accuracy, retention, and learner satisfaction. Taken together, these findings highlight the power of reinforcement learning, particularly when combined with multimodal behavioural data, to drive next-generation intelligent tutoring systems. The system demonstrates promising improvements in adaptivity, efficiency, and generalisation, indicating its potential value for future AI-based language education.

Future work will explore cross-linguistic and cross-domain transfer, real-time affective computing for even finer feedback, and deployment at scale in diverse educational settings. Such directions will further advance both the pedagogical impact and ethical reliability of AI-driven learning platforms. In summary, our work provides a solid foundation for building more adaptive, engaging, and effective AI-powered tutoring systems, paving the way for truly personalised and impactful language education.

Acknowledgements All authors contributed to the study conception and design. All authors read and approved the final manuscript.

Author Contributions All authors contributed to the study conception and design. All authors read and approved the final manuscript.

Funding This work was funded by the Fujian Provincial Department of Education 2024 Educational Reform Project: “Innovation Model Exploration of Piano Cooperative Practice under the ‘Great Ideological and Political Education’ Framework: Integrating Ideological and Political Education with Practical Empowerment” (Project No.: FBJY20240254).

Data Availability The data used to support the findings of this study are available from the corresponding author upon request.

Code Availability The code can be obtained by contacting the corresponding author.

Declarations

Conflict of interest Here are no have no conflicts of interest to declare. All authors have seen and agree with the contents of the manuscript and there is no financial interest to report. We certify that the submission is original work and is not under review at any other publication.

Ethics Approval and Consent to Participate I confirm that all the research meets ethical guidelines and adheres to the legal requirements of the study country.

Consent for Publication I confirm that any participants (or their guardians if unable to give informed consent, or next of kin, if deceased) who may be identifiable through the manuscript (such as a case report), have been given an opportunity to review the final manuscript and have provided written consent to publish.

Human Rights There is no human participate involved in this research. this article manuscript is created from collection of data set.

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